Part I Exam Cheatsheets: Probability and Statistics Theory

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1 Common distributions

Distribution	Mean	Variance	PMF/PDF
Unif $([a,b])$	$\frac{a+b}{2}$	$\frac{(b-a)(b-a+2)}{12}$	$\frac{1}{b-a+1}$
Ber (p)	p	$p(1-p)$	$p^x(1-p)^{1-x}$
Bin (n,p)	np	$np(1-p)$	$\binom{n}{k}p^k(1-p)^{n-k}$
Geom (p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$p(1-p)^{k-1}$
Poiss (λ)	λ	λ	$\frac{\lambda^k e^{-\lambda}}{k!}$
Unif $([a,b])$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{1}{b-a}$
$\mathcal{N}(\mu,\sigma^2)$	μ	σ^2	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
Exp (λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\lambda e^{-\lambda x}$
Beta (α,β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$
Gamma (α,β)	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$

2 MLE and Fisher Information (i.i.d., size n)

Dist	$\ell_n(\theta)$	$\ell_n'(\theta)$	MLE	$I_n(\theta)$
Bern(p)	$\sum X_i \log p + (n-\sum X_i)\log(1-p)$	$\frac{\sum X_i}{p} - \frac{n-\sum X_i}{1-p}$	$\hat{p} = \bar{X}_n$	$\frac{n}{p(1-p)}$
Geom(p)	$n\log p + (\sum X_i - n)\log(1-p)$	$\frac{n}{p} - \frac{\sum X_i - n}{1-p}$	$\hat{p} = \frac{1}{\bar{X}_n}$	$\frac{n}{p^2(1-p)}$
Pois(λ)	$\sum X_i \log\lambda - n\lambda$	$\frac{\sum X_i}{\lambda} - n$	$\hat{\lambda} = \bar{X}_n$	$\frac{n}{\lambda}$
Exp(λ)	$n\log\lambda - \lambda\sum X_i$	$\frac{n}{\lambda} - \sum X_i$	$\hat{\lambda} = \frac{1}{\bar{X}_n}$	$\frac{n}{\lambda^2}$
$\mathcal{N}(\mu,\sigma^2)$	$-\frac{1}{2\sigma^2}\sum(X_i-\mu)^2$	$\frac{\sum(X_i-\mu)}{\sigma^2}$	$\hat{\mu} = \bar{X}_n$	$\frac{n}{\sigma^2}$

3 Conditioning, independence, and counting

3.1 Conditional probability

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \mathbb{P}(A_1)\prod_{i=2}^{n}\mathbb{P}\left(A_i \mid \bigcap_{j=1}^{i-1} A_j\right)$$

$$\mathbb{P}(B) = \sum_i \mathbb{P}(A_i)\,\mathbb{P}(B \mid A_i), \quad \mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\,\mathbb{P}(A_i)}{\sum_j \mathbb{P}(B \mid A_j)\,\mathbb{P}(A_j)}$$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B \mid A)\,\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\,\mathbb{P}(A^c)}$$

3.2 Independence

$$\mathbb{P}\left(\bigcap_{i\in S} A_i\right) = \prod_{i\in S}\mathbb{P}(A_i)$$

- $A \perp B \iff \mathbb{P}(A \mid B) = \mathbb{P}(A)$
- Independence $\not\Rightarrow$ conditional independence
- $X \perp Y \Rightarrow \mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\,\mathbb{E}[h(Y)]$

3.3 Counting

- Permutations: $n!$
- k-permutations: $n!/(n-k)!$
- Combinations: $n!/(k!(n-k)!)$
- Partitions: $n!/(n_1!\cdots n_r!)$

4 Random variables

4.1 Expectation and variance

$$\mathbb{E}[aX+bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y], \quad \mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] = \int \mathbb{E}[X \mid Y=y] f_Y(y)\,dy$$

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \quad \operatorname{Var}(cX) = c^2 \operatorname{Var}(X)$$

$$\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y], \quad \operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)$$

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y]), \quad X \perp Y \Rightarrow \operatorname{Cov}(X,Y) = 0$$

4.2 Correlation and covariance

The *correlation coefficient* measures the linear association between variables:

$$\rho_{XY} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{X_i - \overline{X}_n}{\sigma_X}\right)\left(\frac{Y_i - \overline{Y}_n}{\sigma_Y}\right) = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1,1]$$

Properties of covariance:

- $\operatorname{Cov}(aX+b, Y) = a\operatorname{Cov}(X,Y)$
- $\operatorname{Cov}(X, Y+Z) = \operatorname{Cov}(X,Y) + \operatorname{Cov}(Y,Z)$
- $\operatorname{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \operatorname{Var} X_i + \sum_{\{(i,j)\,|\,i\neq j\}} \operatorname{Cov}(X_i, X_j)$

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4.5 Conditioning rules

$$\mathbb{E}[g(Y)X \mid Y] = g(Y)\,\mathbb{E}[X \mid Y], \quad \operatorname{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - (\mathbb{E}[X \mid Y])^2$$

$$X \perp Y \Rightarrow \mathbb{E}[X \mid Y] = \mathbb{E}[X], \quad \operatorname{Var}(X \mid Y) = \operatorname{Var}(X)$$

4.6 LMS & MAP Estimation

$$\hat{X}_{\text{LMS}} = \mathbb{E}[X \mid Y], \qquad \mathbb{E}[(X - \hat{X}_{\text{LMS}})^2 \mid Y] = \operatorname{Var}(X \mid Y)$$

$$\hat{X}_{\text{MAP}} = \arg\max_x f_{X|Y}(x \mid y)$$

$$\mathbb{E}[(X - \hat{X}_{\text{MAP}})^2 \mid Y] = \operatorname{Var}(X \mid Y) + (\mathbb{E}[X \mid Y] - \hat{X}_{\text{MAP}})^2$$

5 Convergence and limit theorems

5.1 Useful inequalities

Markov: For $X \geq 0$ with $\mathbb{E}[X] > 0$ and $t > 0$, $\mathbb{P}[X \geq t] \leq \mathbb{E}[X]/t$.
Chebyshev: For X with $\mathbb{E}[X] < \infty$ and $t > 0$, $\mathbb{P}[|X - \mathbb{E}[X]| \geq t] \leq (\operatorname{Var} X)/t^2$.
Hoeffding: Given $X_{i\in[n]} \overset{\text{i.i.d.}}{\sim} X$ that are a.s. bounded, i.e., there exist $a < b$ such that $\mathbb{P}[X_i \notin [a,b]] = 0$, then $\mathbb{P}\left[\left|\overline{X}_n - \mathbb{E}[X]\right| \geq \epsilon\right] \leq 2\exp\left(-\frac{2n\epsilon^2}{(b-a)^2}\right)$ for all $\epsilon > 0$.

5.2 Modes of convergence

Let (T_n) be a sequence of r.v.s and T another r.v., all in $\mathbb{R}$.

1. *Convergence almost surely:* $T_n \xrightarrow{\text{a.s.}} T \iff \mathbb{P}[\{\omega \,|\, T_n(\omega) \to T(\omega)\}] = 1$
2. *Convergence in probability:* $T_n \xrightarrow{\mathbb{P}} T \iff \mathbb{P}[|T_n - T| \geq \epsilon] \to 0$ for all $\epsilon > 0$
3. *Convergence in distribution:* $T_n \xrightarrow{\text{d}} T \iff \mathbb{E}[f(T_n)] \to \mathbb{E}[f(T)]$ for all continuous and bounded f

(1) implies (2) implies (3), but (3) implies (2) only if the limit T has a density: $T_n \xrightarrow{\text{d}} T \implies \mathbb{P}[a \leq T_n \leq b] \to \mathbb{P}[a \leq T \leq b]$.
Continuous mapping theorem: Continuous functions preserve limits.

5.3 Limit theorems

Let $X_{i\in[n]} \overset{\text{i.i.d.}}{\sim} X$ with finite mean μ and sample mean $\overline{X}_n$.

- **Strong LLN:** $\overline{X}_n \xrightarrow{\text{a.s.}} \mu$, i.e., $\mathbb{P}\left[\lim_{n\to\infty} \overline{X}_n = \mu\right] = 1$.
- **Weak LLN:** If $\operatorname{Var} X < \infty$, then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$, i.e., $\mathbb{P}\left[\left|\overline{X}_n - \mu\right| \geq \epsilon\right] \to 0$ for all $\epsilon > 0$.

Central limit theorem: If, in addition, $\operatorname{Var} X = \sigma^2 < \infty$, then the sample mean is asymptotically normal, i.e., $\sqrt{n}(\overline{X}_n - \mu) \xrightarrow{\text{d}} \mathcal{N}(0, \sigma^2)$.

Slutsky's theorem: Let (T_n) and (U_n) be sequences of r.v.s such that $T_n \xrightarrow{\text{d}} T$ and $U_n \xrightarrow{\mathbb{P}} u \in \mathbb{R}$. Then

- $T_n + U_n \xrightarrow{\text{d}} T + u$
- $T_n U_n \xrightarrow{\text{d}} Tu$
- $\frac{T_n}{U_n} \xrightarrow{\text{d}} \frac{T}{u}$ if $u \neq 0$.